Cyber security challenges

Examples of cyber security challenges

SingHealth
Defining Tomorrow's Medicine

- Hackers stole the personal particulars of 1.5 million patients after the SingHealth hack during 2018
- 160,000 people including Prime Minister Lee and a few ministers had their outpatient prescriptions stolen as well
- The hacking group was Whitefly, which mostly target Singapore organizations or multinational companies with a presence in Singapore

- MyDoom virus – one of the fastest spreading virus
- Infected a few million computers
- Original author not found, believed to have originated in Russia
- Spreads through email
- Creates a backdoor so that the virus can access hacked computer
- Launches a DDOS attack

Types of Hackers

White hat hackers

- Specializes in penetration testing and other testing methodologies
- Ensures the security of an organization's information systems
- Usually hired by target company

Grey hat hackers

- May violate laws or typical ethical standards
- Does not have the malicious intent typical of a black hat hacker
- Usually for political purposes
- Usually in groups

Black hat hackers

- Violates laws and typical ethical standards
- Motivated by money, espionage, notoriety, and malicious intent
- More daring than white hats
- Hired by individuals/groups

State sponsored hackers

- Hackers appointed by the government to gain information about other countries
- Reports to only government
- Usually well funded

Types of cyber security challenges

Ransomware attack – using a virus to encrypt files and demand a ransom for files to be unencrypted

Phishing attack - an attacker, masquerading as a trusted entity, dupes a victim into opening message

DDoS attack - a malicious attempt to disrupt the normal traffic of a targeted server by overwhelming the target with a flood of Internet traffic

Deepfake – Using AI to impersonate people and spread fake news

SQL Injection - a web security vulnerability that allows an attacker to interfere with the queries that an application makes to its database.

Fileless - a type of malicious software that uses legitimate programs to infect a computer